



Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

College of Computer Studies

ITEC 204 – Data Structures and Algorithms

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Section: BSIT – 2B	Instructor: Ms. Maxyne N. Ignacio

- I. **Objective:** Write a short explanation of what your program is designed to do. (Minimum of 3 sentences)

Answer: My program is designed to manage incident tickets created by automation bots. The program enables the user to add, display, search, remove, and count the number of active incident tickets. It uses the Python list as a linear data structure to store and manage the incident records.

- II. **Program Operations:** Briefly explain how your program performs each operation. (Insert Specific Code Line and Your Explanation)

Add Incident: `tickets.append([incident_id, bot, description])`

- This line allows adding a new incident ticket to the tickets list. This line saves the Incident ID, Bot, and Short Description as a single record in the tickets list. The append() method adds the new record to the end of the list.

Display Incidents: `if ticket[0] == incident_id:`

- This loop checks each record in the tickets list, which allows displaying the list of active incident tickets. As a result, the user can see the list of tickets with their Incident ID, Bot, and Short Description.

Search Incident: `for ticket in tickets:`

- This line checks if the user has entered a valid Incident ID. The result of this condition is used to call the function that displays the required incident ticket or the message "Ticket not found."

Remove Resolved Incident: `tickets.remove(ticket)`

- This line removes the selected incident ticket. First, the loop searches for the needed record in the tickets list, and then it is deleted. This function removes the resolved incident ticket from the active tickets.

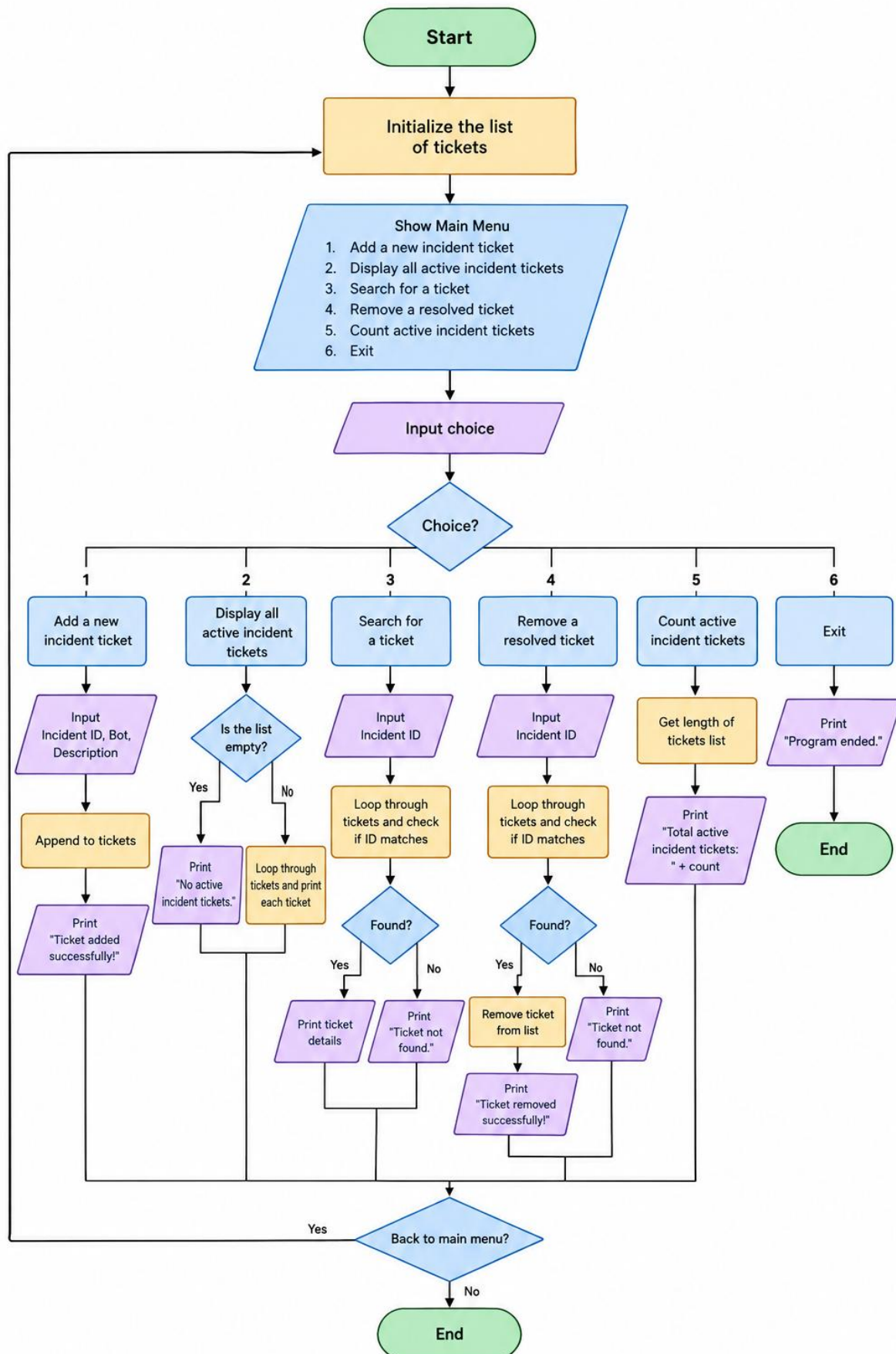
Count Active Incidents: `print("Total active incident tickets:", len(tickets))`

- The len(tickets) function counts the number of records in the tickets list. This function displays the number of active incident tickets. The count of tickets changes when the new record is added or deleted from the list.



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III. **Flowchart:** Create a flowchart showing the overall process and logic of your program.





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- IV. **Conclusion:** Briefly explain what you learned from the activity and how the linear data structure was applied.
- This activity helped me to understand how a linear data structure can be implemented in real IT applications. The activity taught me how to use a Python list to store and manage the incident tickets. The activity also showed how to add, display, search, remove, and count tickets using various list functions. Overall, the activity was important because it deepened my understanding of how a computer program can be used in practice to solve real problems.